Online Web Appendix for

The Power of Disagreement: A Field Experiment to Investigate Human-Algorithm Collaboration in Loan Evaluations

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Appendix A. Summary of Related Works

Table A1 Comparison of Related Studies Regarding Human–Algorithm Collaboration

Research	Information Provided			Main Research Objectives/Focal Outcomes		
	Algorithm Recommendations	Algorithm Performance	Algorithm Rationale (Explanations)	Collaborative Task Performance	Human Responses to Algorithm Recommendations	Mechanisms of Human Responses
Cao et al. (2021)	✓			✓		
Ge et al. (2021)	✓			✓		
Te'eni et al. (2023)	✓			✓		
Luo et al. (2021)	✓			✓		
Dietvorst et al. (2015)	✓				✓	
von Walter et al. (2022)	✓				✓	
Fügener et al. (2021)	✓			✓	✓	
Fügener et al. (2022)	✓			✓	✓	
Lu et al. (2024)	✓			✓	✓	
Sun et al. (2022)	✓			✓	✓	
Allen and Choudhury (2022)	✓			✓	✓	
Castelo et al. (2019)	✓	✓			✓	
Choudhury et al. (2020)	✓	✓		✓	✓	
Ibrahim et al. (2021)	✓	✓		✓	✓	
Lehmann et al. (2022)	✓		✓		✓	
You et al. (2022)	✓	✓	✓		✓	
Bauer et al. (2023)	✓		✓	✓	✓	
Chen et al. (2023)	✓		✓	✓	✓	
Lu and Zhang (2024)	✓		✓	✓	✓	
Liu et al. (2023)	✓			✓	✓	✓
Our study	✓		✓	✓	✓	✓

Appendix B. Supplementary Information on Research Context and Experiment Implementation

B1. Data Collection and Usage

All loan applicants were explicitly asked to authorize the firm to access their data as a prerequisite for submitting their loan applications, and they were also guaranteed that the data would be solely used for credit risk assessment and would not be disclosed to or shared with third parties by any means. Because a majority of applicants were in urgent financial need, over 80% consented to this data collection and usage policy.

The IT department of the focal firm handled the initial data cleaning and preparation tasks. We then completed the remaining tasks, including feature construction, model training, and model prediction, with the help of the IT department. The algorithms were eventually implemented by the focal firm. During the whole process, all identifiable information about the loan applicants (e.g., real names, national IDs, and cellphone numbers) was encrypted by the firm and invisible to us.

We endeavored to design the experiment with full ethical consideration and comply with related regulations. The practical motivation for the experiment was to improve lending efficiency and effectiveness, which ultimately aligned with the goal of financial inclusion in the country and the social welfare of loan applicants. The introduction of the algorithms was also aimed at lowering the burden of human evaluators in the long run. Our empirical results confirmed that the experiment did not diminish the benefits nor expose risks to loan applicants or human evaluators. The findings of the experiment provided implications for improving the welfare of both loan applicants (e.g., rejecting overly stretching applicants and lending to underserved but creditworthy ones) and human evaluators (e.g., where and how to pay attention when working with algorithms). In addition, the usage of personal data, the development of models, and the implementation of financial technologies by the focal firm were all regulated in the focal country. [1]

B2. Human Evaluator Training and Algorithmic Model Training

Prior to the experiment, the firm conducted a training course to familiarize the human evaluators with the newly designed scenarios and available features of each application. Specifically, the firm randomly selected approved applications in June 2017. For each application, the firm obtained the final repayment behavior (i.e., default or not), the 12 basic features, and the 64 additional features. All human evaluators were first instructed on the definitions and measurements of these features and then went through eight rounds to reach stable evaluation criteria and accuracy. In each round, the firm asked them to first make lending decisions on 100 randomly chosen loans based on

[1] There was no explicit IRB under effect for our field experiment, primarily due to the lack of an academic ethical review system in the focal university (as well as in the focal country) of the author who mainly designed and helped implement the algorithms back at that time. However, the author who physically conducted the experiment indeed followed the basic rules for experiments involving human subjects, including providing fair treatment, minimizing harm, ensuring information transparency, obtaining explicit consent, etc.

available features (which depended on the assigned experimental groups, as described in our two-step experimental design) and then revealed the actual repayment performance. This training approach helped rule out interference caused by the lack of knowledge or decision-making immaturity in our experiment. The evaluators were told that algorithms usually excel in decision-making, but they did not know much about applied ML algorithms beyond what was trained at the focal firm. The firm had not implemented any lending algorithms prior to our experiment, and the human evaluators had no prior experience working with other lending algorithms.

Regarding the selection and training of the loan evaluation algorithms, we followed Lu et al. (2023) to do feature generation and algorithmic model training. Specifically, we attempted multiple ML algorithms, including logistic regression, k-nearest neighbor, support vector machine, multi-level perceptron, random forest, and XGBoost. XGBoost models achieved the best performance under both limited- and rich-information scenarios (overall accuracy of 0.867 and 0.924, respectively). For all models, we employed standard operationalization, including hyperparameter tuning, ten-fold cross-validation, and out-of-sample prediction techniques. The XGBoost models (as well as other models) were trained on historical loan cases approved by human evaluators. This approach is not only practically viable but also widely used in many credit risk prediction studies (e.g., Fu et al. 2021, Zhou et al. 2021). Although this approach may introduce inherent biases, we believe these biases were evenly distributed across all experimental groups, thus minimizing their impact on our empirical findings. We encourage future studies to find better ways to mitigate biases in model training.

B3. Experimental Design and Randomization Checks

Below is the message we used to describe the AI algorithms to the participating human evaluators in our experiment: *"We are excited to introduce a new tool to assist your loan evaluation process. We have integrated an advanced machine-learning-based AI algorithm to predict loan default rates and provide recommendations for your final lending decision-making. Overall, AI algorithms are highly proficient at making accurate predictions, which, hopefully, can help you enhance the quality of your evaluations. It's important to note that you remain the ultimate decision maker, and the AI algorithms will be a valuable reference to help benefit your loan evaluation process."*

Because of the training course and the clear guidance, we are confident that human evaluators were motivated to seriously consider and utilize algorithmic recommendations in our experiment. However, we still conducted multiple manipulation and randomization checks to make sure the experiment was implemented as expected (see Appendix Table B1).

It is worth noting that during the experiment, human evaluators were instructed to spend approximately two hours each day reviewing their assigned applications during the experiment. Each evaluator reviewed around 400 applications (approximately 6,000 total applications divided among 15 evaluators in each group, referring to Table B1) over 10 workdays (i.e., two weeks), averaging 40 applications per day. Evaluators spent approximately 2

to 3 minutes reviewing each application. While there may be some variance in evaluation time across applications and evaluators, this variance is generally small. This is because all evaluators were required to complete their tasks within the specified two-hour window, and none finished significantly earlier, ensuring consistency in the total time spent on the evaluation task.

Table B1 Randomization Check

	Loan attributes				
Group	Total # of applications	Loan amount (US$)	Loan term (months)	Loan interest rate (%)	Loan purpose
1. L+ND	5,892	470.478	5.425	14.059	0.415
		(68.199)	(1.520)	(1.089)	(0.493)
2. R+ND	6,010	470.723	5.412	14.044	0.423
		(65.471)	(1.514)	(1.089)	(0.494)
3. L+D	5,895	469.053	5.411	14.050	0.417
		(66.397)	(1.512)	(1.094)	(0.493)
4. R+D	6,012	471.403	5.411	14.072	0.415
		(63.341)	(1.516)	(1.123)	(0.493)
	Applicant characteristics				
Group	Age	Gender	Education level	Monthly income level	Living city DPI (US$)
1. L+ND	24.585	0.214	4.216	3.688	6,358.104
	(4.042)	(0.410)	(0.690)	(1.892)	(2,390.265)
2. R+ND	24.621	0.213	4.221	3.664	6,416.840
	(4.049)	(0.409)	(0.667)	(1.868)	(2,431.346)
3. L+D	24.681	0.213	4.226	3.651	6,385.161
	(4.085)	(0.410)	(0.671)	(1.886)	(2,426.075)
4. R+D	24.618	0.211	4.261	3.709	6,408.834
	(4.112)	(0.408)	(0.683)	(1.912)	(2,422.113)
	Human evaluator objective characteristics				
Group	#Evaluators	Gender	Education level	Tenure (months)	Previous accuracy
1. L+ND	16	0.938	4.500	2.438	2.125
		(0.250)	(0.614)	(1.020)	(0.885)
2. R+ND	15	0.600	4.465	2.669	1.933
		(0.463)	(0.556)	(1.275)	(0.799)
3. L+D	15	0.600	4.400	2.600	1.867
		(0.482)	(0.632)	(1.121)	(0.915)
4. R+D	16	0.938	4.437	2.376	2.188
		(0.250)	(0.712)	(0.890)	(0.750)

[a] Standard deviations are reported in parentheses. / Between-group *F*-tests generally indicate no significant discrepancies among groups for variables.
[b] L (R) = limited (rich) information; ND (D) = without (with) disclosure of important features used by algorithms.
[c] Loan purpose: 1 = consumption, 0 = others (e.g., used for emergent events).
[d] Gender (for both loan applicants and human evaluators): 1 = female, 0 = male. Education level (for both loan applicants and human evaluators): 1 = middle school or below, 2 = vocational school, 3 = high school, 4 = technical school, 5 = undergraduate, 6 = graduate or above. Monthly income level: 1 = US$150 or below, 2 = US$150–US$300, ..., 9 = US$1,200 or above.
[e] Tenure: 1 = not longer than 6 months, 2 = 6–12 months, 3 = 13–18 months, 4 = longer than 18 months. Previous (decision) accuracy: 1 = low (default rate > 15%), 2 = medium (10% < default rate < 15%), 3 = high (default rate < 10%).

B4. Loan Performance Imputation

As mentioned in the main text, in order to construct the measurement of "good decision", we had to approximate all denied applicants' counterfactual repayment behavior. To do that, we obtained additional information from the

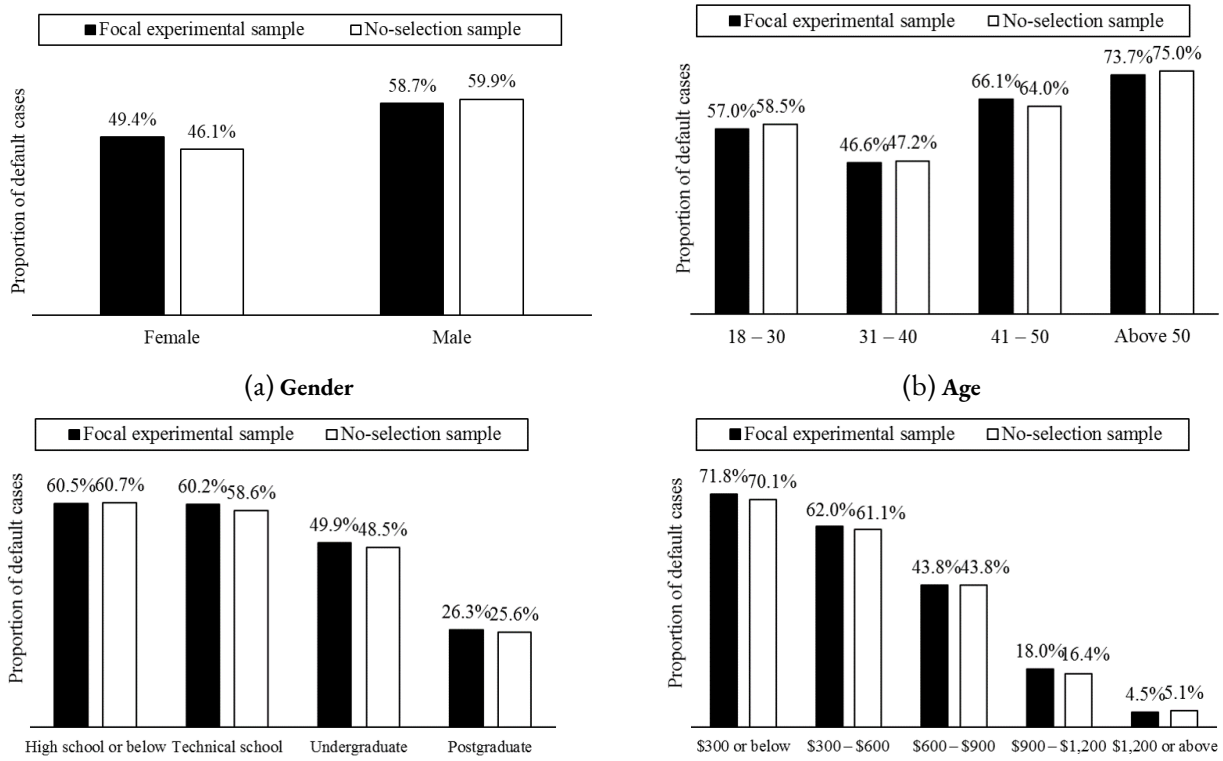
same firm with another independent experiment design. This alternative data set included similar applicants (i.e., within the same applicant pool) who had been randomly selected for another experiment run by the same firm for other business purposes. The firm implemented this experiment (named "alternative experiment" hereafter) from December 2 to 22, 2017, and there was no overlap in applicants between the two experiments. In this alternative experiment, the firm approved all applications without loan evaluation or screening. Consequently, we were able to observe the actual repayment behavior (i.e., default or not) of all applicants. We named this alternative sample the "no-selection" sample, which covered the entire distribution of the applicant population. In total, the no-selection sample consisted of 7,726 applications, containing the same feature sets (i.e., both basic and additional features) and repayment behavior data.

We leveraged this no-selection sample to approximate the counterfactual repayment behavior of the denied applications in the focal experiment with prediction techniques. We used the no-selection sample as a training data set to train an XGBoost model, and we applied the model to the focal experimental sample as an out-of-sample prediction set in order to impute the repayment behavior of denied applications.[2] To validate the effectiveness and reliability of the above XGBoost-based imputation approach, we conducted several checks on the descriptive statistics and presented the evidence from different dimensions. *First*, the in-sample prediction for the no-selection sample was satisfactory, showing an accuracy rate and AUC of 0.981 and 0.984, respectively. *Second*, because we could also impute the approved applications in the focal experimental sample, we compared the predicted and actual performance of these approved applications, finding that the prediction accuracy and AUC were 0.976 and 0.980, respectively.[3] *Third*, we compared the proportion of defaulted loans between the focal experimental sample and the no-selection sample based on major applicant objective features and reported the comparison results in Appendix Figure B1. We observed no differences between the two samples within any of the sub-groups (e.g., among female applicants or among applicants with a postgraduate degree, etc.). Given that both samples were randomly selected from the same applicant pool, the distributions of applicant features were supposed to be the same. Thus, the consistency of the predicted default labels within each demographic sub-group suggested that our imputation was not biased by features. *Finally*, we checked the prediction performance for each experimental group. As shown in Appendix Figure B2, again, we observed no significant difference across groups. This implies that even if our imputation was not perfect, the unobserved errors were almost identical across the four experimental groups, which in turn would not bias our subsequent empirical findings. Based on the aforementioned evidence, we were confident about the performance imputation approach. Therefore, we used the true performance of approved

[2] We first employed an XGBoost model with all 76 features on the training set to predict applications' default behavior. We then imputed the repayment performance for the denied applications in our focal experimental sample. The prediction utilized all 76 features. We operationalized the prediction task with standard procedures such as hyper-parameter tuning, 10-fold cross-validation, and the oversampling method.

[3] It is worth noting that the extremely high accuracy was due to the usage of an unbiased training sample (i.e., including all applications). As revealed in Lu et al. (2023), such an ideal sample could significantly improve the performance of machine learning models. However, it is not realistic in practice, and thus, in our focal experiment on human-algorithm collaboration, we trained the two algorithms only with data from approved applicants.

loans and the imputed performance of denied applications to identify good decisions. The results were consistent when we used the imputed performance of both approved and denied applications.



[a] The statistics on the focal experimental sample came from the XGBoost prediction model, which was trained by the no-selection sample to impute loan performance. The accuracy rates of these models were all over 0.97. The empirical results also suggested that the imputation approach was reasonable and reliable because the full sample and the no-selection sample share an almost identical distribution of defaulted loans across multiple features.

Figure B1 Proportion of Defaulted Loans by Feature Values with the Full Sample (Based on Imputed Loan Performance)

		Predicted	
		Non-default	*Default*
Actual	*Non-default*	2,218	69
	Default	0	290

Accuracy: 0.973
Precision: 1.000
Recall: 0.970

(a) **Group 1 (L+ND)**

		Predicted	
		Non-default	*Default*
Actual	*Non-default*	2,391	61
	Default	1	167

Accuracy: 0.976
Precision: 0.999
Recall: 0.975

(b) **Group 2 (R+ND)**

		Predicted	
		Non-default	*Default*
Actual	*Non-default*	2,248	73
	Default	1	276

Accuracy: 0.972
Precision: 0.999
Recall: 0.969

(c) **Group 3 (L+D)**

		Predicted	
		Non-default	*Default*
Actual	*Non-default*	2,477	41
	Default	1	99

Accuracy: 0.984
Precision: 0.999
Recall: 0.983

(d) **Group 4 (R+D)**

[a] The statistics were computed based on the XGBoost prediction model. Different from the XGBoost prediction models used for our focal experiment, this prediction model was used for loan performance imputation.

Figure B2 Comparison of Prediction Performance across Groups for the Loan Performance Imputation Task (Based on Approved Applications in the Full Sample)

Appendix C. Comparison of Ratios of Good Decisions

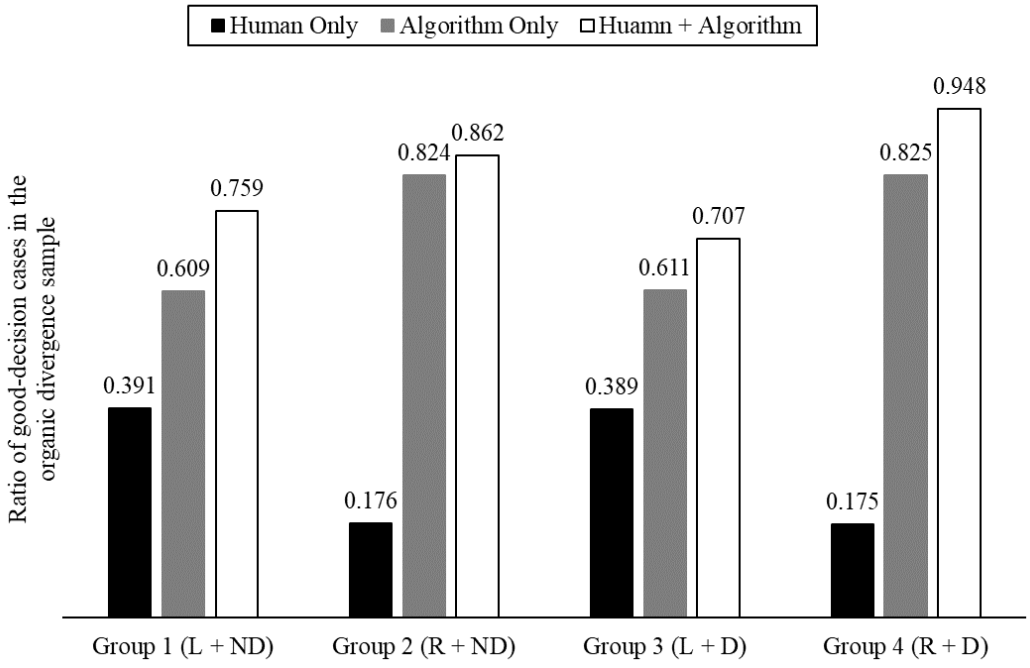


Figure C1 Ratio of Good Decisions (Organic-divergence Sample)

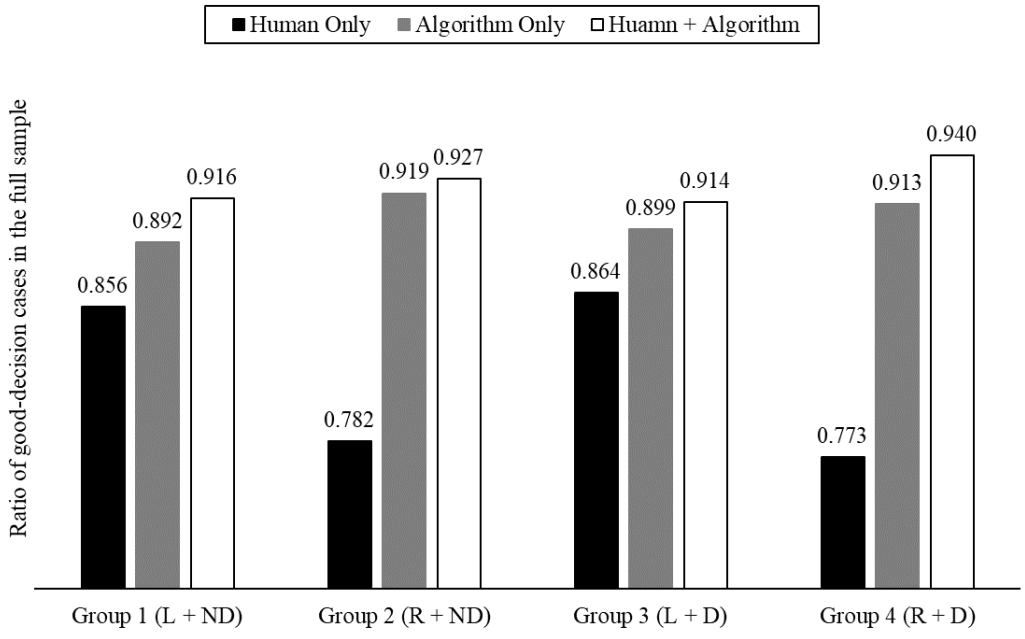


Figure C2 Ratio of Good Decisions (Full Sample)

Appendix D. Supplementary Analysis: Disagreement and Collaborative Value

We examined the relationship between disagreement and the two performance metrics. Specifically, we employed an application-level linear regression model. The analysis used the full sample and included group-level fixed effects to control for the unique setting of each group. Table D1 reports the estimated coefficients. First, Columns 1 and 2 show that organic divergence was negatively associated with GoodDecision but positively associated with DecisionAugmentation. One possibility is that the organic-divergence samples were naturally complicated and more likely to receive bad decisions than the rest. However, the positive coefficient in Column 2 suggests that when human evaluators disagreed with algorithmic recommendations in the first step, no matter what happened later (i.e., accept algorithms or not), on average, they were more likely to augment algorithm-only decisions rather than to undermine algorithm-only decisions. Generally speaking, human evaluators were careful and knowledgeable enough to utilize algorithmic recommendations. This was also confirmed by the positive and larger coefficient of `CollaborativeDisagree` in the last column. The relationship between collaborative disagreement and good decision was negative in Column 3, which was different from the positive coefficient in the last column. Considering that there existed systematic differences between applications with or without collaborative disagreements (i.e., applications with collaborative disagreements might be more complicated and uncommon), it is possible that the absolute performance was lower when ColaborativeDisagree $= 1$. However, we could still observe a direct collaborative value via DecisionAugmentation.

Table D1 Organic Divergence, Disagreement, and Collaborative Performance

D.V.	`GoodDecision`	`Augmentation`	`GoodDecision`	`Augmentation`
`OrganicDiverge`	-0.115***	0.099***	-0.084***	-0.0006*
	(0.006)	(0.007)	(0.006)	(0.0003)
`CollaborativeDisagree`			-0.137***	0.448***
			(0.015)	(0.028)
Group Fixed Effects	✓	✓	✓	✓
Observations	23,809	23,809	23,809	23,809
Adjusted R^2	0.030	0.038	0.039	0.194

[a] Robust standard errors are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. This table uses the full sample.

Appendix E. Supplementary Questionnaire Design and Items

To understand human evaluators' attitudes toward and perceptions of the firm's ML algorithms and (rich) borrower information, we devised a questionnaire of 23 items for evaluators to complete immediately after the experiment. The questionnaire also touched upon their general thoughts and beliefs. The focal company incorporated the questionnaire description and items into their loan evaluation system and launched the online questionnaire as soon as human evaluators finished all lending decision tasks in our experiment. Every human evaluator fully completed the questionnaire, which used a 7-point Likert scale ranging from "strongly disagree" to "strongly agree."

The questionnaire items were developed based on our review of existing human-algorithm collaboration literature and the manager's business insights. The items fall into three categories: perceptions of the firm's loan evaluation algorithms, perceptions of the available (rich) borrower information, and general thoughts and beliefs. To avoid potential confusion, human evaluators in different experimental groups received distinct versions of the questionnaire. That is, some questions were only presented to groups with algorithm rationale and/or rich information. Table E1 displays the detailed questionnaire description and items. We controlled all of them in our models (based on the availability of each item across various experimental groups), and the results on the questionnaire items are available upon request.

Table E1 Questionnaire Description and Items

For each of the following questions, please choose the option that best describes your thoughts or experience:
1 = strongly disagree, 2 = somewhat disagree, 3 = slightly disagree, 4 = neutral, 5 = slightly agree, 6 = somewhat agree, 7 = strongly agree.

ID	Question item	Mean value
Perception of the firm's loan evaluation algorithms		
Q1	Overall, I think the firm's machine learning (ML) algorithm is *not accurate* enough.	3.817
Q2	At work, I tend to *trust* the decision recommendations provided by the firm's ML algorithm.	4.046
Q3	At work, I tend to make loan evaluations by leveraging *both my own experience and the algorithmic recommendations*.	3.956
Q4	Overall, I think introducing ML algorithms into my work has *increased the coordination and decision-making costs*.	2.912
Q5	Overall, I do *not want to adopt* the algorithmic recommendations.	4.108
Q6	Overall, I think there is *no need* to introduce any AI algorithms in my work as I believe my evaluations are accurate enough.	3.144
Q7	Overall, I think the *explanations* for the algorithmic recommendations are *vague and unclear*.	4.645
Q8	Overall, I have *difficult in understanding* the algorithm rationale.	4.546
Perception of the available (rich) borrower information		
Q9	Overall, I do *not want to consider* the rich information of borrowers in each application.	3.030
Q10	Overall, I think there is *no need* to use rich information of borrowers as I believe my evaluations based on existing limited basic data are accurate enough.	3.391
Q11	Overall, I think the rich information of borrowers has *trivial value* for loan evaluations.	3.270
Q12	Overall, I think the rich information of borrowers has actually interfered with my own loan evaluation practice.	3.612
Q13	Overall, I have *difficult in understanding* the meaning of the rich information of borrowers.	3.429
Q14	Overall, I do *not know how to effectively use* the rich information of borrowers.	3.647
Q15	Usually, for each borrower, I evaluate the credit risk based on *only a few* (less than or equal to five) borrower features.	5.308
Q16	Overall, I manage to find ways to *take full advantage* of the rich information of borrowers in loan evaluations.	3.179
General thoughts and beliefs		
Q17	Overall, I think I have *a good understanding* of the basic knowledge, working principles, and applications of ML algorithms.	2.549
Q18	Overall, I have *a skeptical and cautious attitude* toward all new ML and AI technologies.	4.399
Q19	Overall, I have *a skeptical and cautious attitude* toward the use of rich information.	3.739
Q20	Overall, I am worried that ML and AI algorithms will one day *replace* my job.	4.517
Q21	Overall, I continuously *reflect on my own experiences* to improve my decision-making capability at work.	3.395
Q22	Overall, I constantly *seek advice* from managers and colleagues to improve my decision-making capability at work.	2.734
Q23	Overall, I have *strong confidence* in my own working capabilities.	4.693

[a] Human evaluators in different experimental groups were presented with distinct versions of the questionnaire. Specifically, Q7 and Q8 (about algorithm rationale) were given only to Groups 3 and 4, while Q9 through Q16 and Q19 (about rich information) were exclusive to Groups 2 and 4. All other questions were common to all groups.

[b] The actual questionnaire presented to human evaluators was in the language of the focal country.

Appendix F. Supplementary Details of Variable Construction

F1. Decision-rule Heterogeneity

To capture the heterogeneity in algorithms' and human evaluators' decision rules, we developed and applied a two-step variable construction procedure:

- Step 1: fit linear regression models to simulate the decision-making rules for each evaluator or algorithm;
- Step 2: identify the significant features in each linear model, record the signs of the corresponding coefficients, and calculate Consistency and Conflict.

We next elaborated on the two steps in detail below.

Step 1: In order to make the comparison of decision-making models between each human evaluator and algorithm both tractable and feasible, we considered linear regression models to simulate both parts' decision-making rules. Although we acknowledged that the actual decision-making rules might not follow a linear pattern, we chose this approach for three reasons: (1) applying the same model to algorithms and human evaluators allows us to make comparisons directly; (2) linear models are more easily implemented and more interpretable than sophisticated algorithmic models such as XGboost; (3) human evaluators tend to assume linear interpretations between features and the target (i.e., credit risk). Specifically, we fitted the linear regression models with two specifications: one with a limited feature set (i.e., eight basic features suggested by the firm) and the other with a rich feature set. [4]. Because the rich feature set contained 76 different features in total, including all of them in a simple linear regression may incur issues of collinearity and low degree of freedom. Thus, to approximate the decision rules under the rich-information scenario, we included only basic features and four additional features: those that had been identified with the highest feature importance and disclosed to human evaluators in Group 4. As a result, we fitted one linear regression model using 8 features for the algorithm under the limited-information scenario and another model using 12 features under the rich-information scenario. We also fitted one model for each evaluator, using 8 or 12 features depending on their group.[5] The dependent variable of each fitted linear regression model was the algorithm's or the evaluator's initial decision without any collaboration. We estimated the coefficients using all applications reviewed by an algorithm or a human evaluator in our experiment. Table F1 presents the estimated coefficients of the two ML algorithms and two sampled evaluators. The coefficients indicated the association between each variable and the initial lending decision.

Step 2: Based on the above approximation, we identified the signs and significant levels of all model features. We measured Consistency (between an evaluator's decision model and an algorithm's decision model within the same

[4] As suggested by the focal firm, we did not include four loan-level features in the models, which were historical overdue in other platforms, loan amount, interest rate, and loan term. The main reason to exclude them was that they did not have large variations among applications. Thus, human evaluators always ignore them when reviewing applications. We did not observe a significant role of any of them in the linear regression models either.

[5] Under the rich-information scenario, we did not argue that the 12 features were necessarily the "real" decision factors that human evaluators relied on during the decision-making process, but they helped to reflect human evaluators' attitudes toward all potentially related factors.

Table F1 Fitted Decision Models

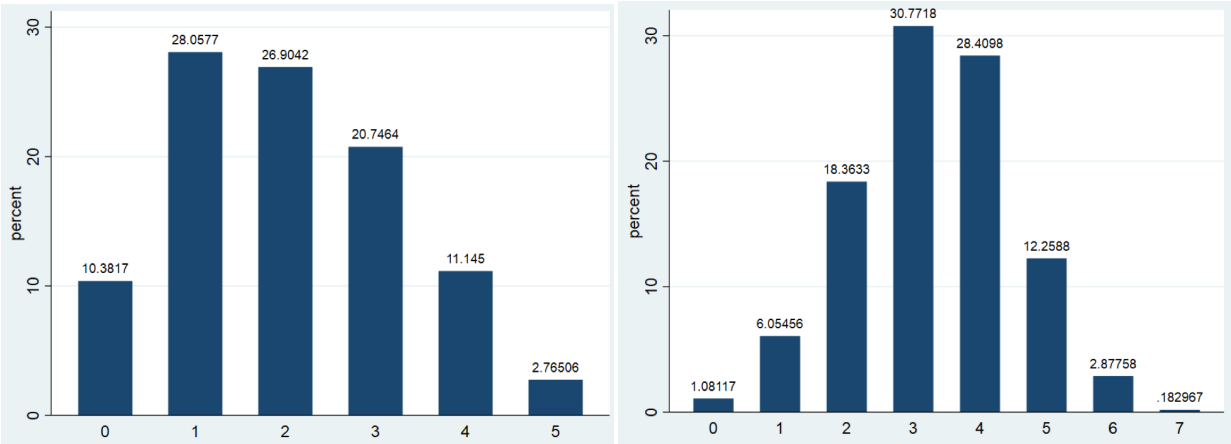
	RichInfo = 0		RichInfo = 1	
Decision Model	ML Algorithm	Evaluator E1	ML Algorithm	Evaluator E2
Loan purpose	-0.027*** (0.008)	-0.021 (0.046)	-0.008 (0.007)	0.052 (0.045)
Gender	0.064*** (0.010)	0.092 (0.057)	0.089*** (0.009)	0.034 (0.054)
Age	0.021*** (0.001)	0.018*** (0.006)	0.019*** (0.001)	0.025*** (0.006)
Living city DPI	0.056*** (0.002)	0.052*** (0.009)	0.040*** (0.002)	0.038*** (0.009)
Monthly income level	0.046*** (0.003)	0.054*** (0.018)	0.036*** (0.003)	0.008 (0.016)
Loan-to-income ratio	-0.013*** (0.005)	0.057* (0.030)	-0.014*** (0.005)	-0.044 (0.028)
Home ownership	0.040*** (0.011)	0.029 (0.063)	0.031*** (0.010)	0.045 (0.061)
Education level	0.043*** (0.006)	0.028 (0.033)	0.014** (0.005)	-0.022 (0.033)
Average amount of game card			-0.005*** (0.000)	-0.005*** (0.002)
# of outgoing contacts			0.003*** (0.001)	-0.004 (0.003)
# of weekly office visits			0.022*** (0.001)	0.022*** (0.004)
# of weekly commercial place visits			0.020*** (0.001)	0.006 (0.006)
Observations	11,787	402	12,022	394
Adjusted R^2	0.197	0.144	0.337	0.240

[a] Robust standard errors are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. This table uses the applications reviewed by each evaluator or algorithm.

experimental group) by the number of features that presented the same directions *and* were statistically significant in both models. Consistent directions ensure humans and algorithms thought in the same way, while significant signs ensure that these decision factors were actually important. Taking Evaluators E1 and E2 as examples (i.e., Columns 2 and 4 in Table F1), their value of Consistency are 3 (i.e., age, living city DPI, and monthly income level) and 4 (i.e., age, living city DPI, average amount of game card, and the number of visits to office place per week). On the other hand, we measured Conflict by the number of features that were of opposite directions *and* statistically significant in both models. The two sampled evaluators' values of Conflict are 1 (i.e., loan to income ratio) and 0. We developed Consistency to measure the extent to which humans and algorithms could seriously agree on key factors and developed Conflict to measure the extent to which humans and algorithms could seriously disagree on key factors. In the analysis, because we always controlled experimental groups using group-fixed effects, their scale was not a concern. Our main results were robust if we measured them in ratios (e.g., consistent factors/total factors).

F2. Perceived Algorithm Quality

As we demonstrated in Figure 2b, the firm disclosed algorithm rationale to the evaluators in Groups 3 and 4. We revealed the names of the important features and the signs of their effects on loan performance (using a Shapley value analysis). We also displayed the focal application's values of the corresponding features and the average value of (historical) non-defaulters. To be more specific, a positive (negative) sign of the feature suggested that a higher value of that feature would lead to a higher (lower) probability of default. Based on this experimental setting, we first defined three indicators: $\{\mathsf{LowValue}\} = 1(0)$ if the value of the focal application was lower (higher) than the average value of non-defaulters; $\{\mathsf{Sign}\} = 1(0)$ if the sign of the feature was positive (negative); and



(a) **Limited-information Scenario** (b) **Rich-information Scenario**

Figure F1 Distribution of Algorithm Self-Contradiction Levels

$\{\mathsf{AlgorithmicApproval}\} = 1(0)$ if the algorithm recommended approving (denying) the application. Thus, a feature identified by the algorithm was self-contradictory if it satisfied any of the four conditions specified in Table F2. Intuitively, algorithm self-contradiction happened when the algorithm's logic and the value of a feature point in one direction, but the algorithm actually recommended the opposite decision. For example, in the sampled application presented in Figure 2b, $\mathsf{Contradiction} = 3$ (i.e., living city DPI, the weekly number of visits to commercial places, and education level). Based on this, it was reasonable for human evaluators to believe the ML algorithms could not fit the particular application well.

Table F2 Conditions of Self-Contradictory Features

Conditions	{LowValue}	{Sign}	{AlgorithmicApproval}
A	1	1	0
B	1	0	1
C	0	1	1
D	0	0	0

Because Groups 1 and 2 did not disclose algorithm rationale, human evaluators did not have any clue to infer Contradiction. We thus defined its value to be zero. We reported the distribution of Contradiction within Groups 3 and 4, respectively, in Figures F1a and F1b. Both distributions were close to normal distributions.

F3. Robustness Check: Alternative Measures of Independent Variables

To investigate the robustness of our key independent variables, we applied alternative ways to reconstruct Contradiction, Consistency, and Conflict. First, we designed a strict version of the original Contradiction measure and named it ContradictionStrict. Different from the process described in Appendix F2, we required the value of a variable to be beyond one standard error of the mean value to be considered as low value or high value, whereas previously Contradiction was just below or above the mean value. Following the same logic, algorithm

self-contradiction happened when the algorithm's logic and the value of a feature pointed to one direction, but the algorithm actually recommended the opposite decision. We called this a strict version of Contradiction because it required a sharp contrast between the value in each application and the mean value. Row 1 in Table F3 reports the mean values of it across four groups, indicating that Group 4 (rich information plus disclosure of algorithm rationale) has the highest level of ContradictionStrict.

Second, we developed two measures to reflect the differences between human evaluators and algorithms in loan evaluation. Building upon the approach described in Appendix F1, we used the vector of coefficients estimated in the fitted decision models to approximate the decision-making rules of humans and algorithms. For example, Column 1 in Table F1 suggests that the algorithm applied a vector of (-0.027, 0.064, 0.021, 0.056, 0.046, -0.013, 0.040, 0.043) in its evaluation under the limited-information scenario. Column 2 suggests that Evaluator E1 applied a vector of (-0.021, 0.092, 0.018, 0.052, 0.054, 0.057, 0.029, 0.028). Then, we calculated both the Euclidean distance and the Cosine distance between the two vectors to approximate the differences between their decision-making rules (because these two distance measures might capture different aspects of the differences), denoted as EuclideanDist and CosineDist, respectively. We reported the mean values of these two measures across four groups in Rows 2 and 3 in Table F3.

Third, we developed a variable called ConsiderationSet to measure the number of factors that were significant in the fitted models to compensate for the overlook of significance in the measurement of Euclidean and Cosine distances. ConsiderationSet thus suggests how firm and consistent each human evaluator was in loan evaluation. For example, the value of ConsiderationSet is four for evaluator E1 in Table F1. We reported the mean values of it across four groups in Row 4 in Table F3. The ConsiderationSet of evaluators under the rich-information scenarios was significantly larger than that of evaluators under the limited-information scenarios, suggesting evaluators indeed utilized the rich information in those two groups.

Table F3 Statistics of Alternative Measures

Group	L+ND	R+ND	L+D	R+D
Contradiction (strict version)	0	0	0.934	1.749
Euclidean distance	0.142	0.133	0.116	0.127
Cosine distance	0.503	0.535	0.405	0.505
Consideration set	3.138	4.837	3.806	5.368

After we created these alternative measures, we first examined whether they make sense and to what extent they are related to the original measures. We checked the correlation among these variables plus two subjective variables from the evaluator questionnaire. As shown in Table F4, the original Contradiction measure and the strict version ContradictionStrict are highly correlated (0.819). The two new distance measures are highly correlated too (0.792), and both of them are positively correlated with Conflict and negatively correlated with Consistency. Interestingly, ConsiderationSet is highly correlated with Consistency (0.936), meaning that human evaluators with a larger set

Table F4 Correlation Table

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
(1) No Adoption Intention (Q5)	1								
(2) Lack of Usefulness (Q6)	0.206	1							
(3) Consistency	-0.283	0.045	1						
(4) Conflict	-0.157	-0.030	-0.319	1					
(5) EuclideanDist	-0.073	-0.022	-0.072	0.377	1				
(6) CosineDist	-0.084	-0.033	-0.316	0.425	0.792	1			
(7) ConsiderationSet	-0.365	-0.032	0.936	0.016	0.076	-0.166	1		
(8) Contradiction	-0.160	-0.025	0.218	0.038	-0.106	-0.044	0.228	1	
(9) ContradictionStrict	-0.155	-0.001	0.188	0.031	-0.084	-0.030	0.196	0.819	1

of firm and consistent decision factors are also likely to share a larger set of similar factors with the algorithms. Two possible explanations include that algorithms used every factor in a significant way and that it was not hard, for either humans or algorithms, to find the reasonable impact direction of each factor. Another observation is that all these variables are not closely related to the intention to adopt algorithmic recommendations or the perceived usefulness of algorithms, suggesting that they capture something else beyond intention and attitude.

We then replaced the original variables with alternative measures separately first and altogether later. Table F5 reported the results. In all cases, the baseline impact of disagreement is still positive and significant, indicating its value in human-algorithm collaboration. Disagreement was always helpful for decision augmentation when it was triggered by Contradiction, regardless of whether it was measured in the original or strict versions. The moderation effect of Euclidean or Cosine distance is positive, which is different from the impact of Conflict. One possible explanation is that ConsiderationSet captures the previous impacts of Consistency and Conflict because it reflects how firm and consistent each evaluator was in evaluating loan applications. ConsiderationSet is highly correlated to Consistency and also exhibits a negative moderation effect on disagreement. When human evaluators have a larger set of factors to consider in their decision-making, their disagreement with algorithmic recommendations is more likely to hurt collaborative value. Given the results in our main model and the results here, we believe that a mature, long-lasting, fixed decision-making pattern hinders human-algorithm collaboration, possibly due to overconfidence or insistence. In contrast, human evaluators who are flexible (or unstable) with decision-making factors are more likely to disagree with algorithms in a beneficial way.

F4. Robustness Check: Refined Measures of Decision Augmentation

Decision augmentation can occur in two scenarios: type I decision augmentation occurs when algorithms incorrectly recommend denial of a good application, but human evaluators override this recommendation; type II decision augmentation occurs when algorithms mistakenly recommend approval of a bad application, but human evaluators do not follow this recommendation. We distinguished these two decision augmentation types and showed the performance comparison results from the field experiment in Figure F2, which demonstrates a similar pattern to the one in our main findings (i.e., Figure 3b). We further replicated the decision augmentation model using the

Table F5 Robustness Checks with Alternative Measures of Key Independent Variables

	Decision Augmentation				
Model	Strict contradiction measure	Euclidean distance	Cosine distance	Euclidean distance + Strict contradiction	Cosine distance + Strict contradiction
CollaborativeDisagree × Consistency	-0.022				
	(0.019)				
CollaborativeDisagree × Conflict	-0.007				
	(0.047)				
CollaborativeDisagree × EuclideanDist		1.429**		1.212**	
		(0.581)		(0.586)	
CollaborativeDisagree × CosineDist			0.141*		0.158*
			(0.081)		(0.083)
CollaborativeDisagree × ConsiderationSet		-0.058***	-0.043**	-0.036**	-0.023
		(0.018)	(0.018)	(0.018)	(0.019)
CollaborativeDisagree × Contradiction		0.160***	0.154***		
		(0.013)	(0.013)		
CollaborativeDisagree × ContradictionStrict	0.194***			0.202***	0.195***
	(0.021)			(0.021)	(0.021)
CollaborativeDisagree	0.404***	0.278***	0.343***	0.307***	0.337***
	(0.084)	(0.103)	(0.096)	(0.104)	(0.098)
Consistency	0.000				
	(0.005)				
Conflict	0.027**				
	(0.014)				
EuclideanDist		-0.064		-0.029	
		(0.096)		(0.098)	
CosineDist			-0.025*		-0.027*
			(0.014)		(0.014)
ConsiderationSet		0.001	-0.000	-0.003	-0.004
		(0.004)	(0.004)	(0.004)	(0.004)
Contradiction		0.008*	0.009**		
		(0.005)	(0.005)		
ContradictionStrict	-0.020***			-0.022***	-0.020***
	(0.005)			(0.005)	(0.005)
Human Characteristics	✓	✓	✓	✓	✓
Loan Features	✓	✓	✓	✓	✓
Group Fixed Effects	✓	✓	✓	✓	✓
Observations	4,451	4,451	4,451	4,451	4,451
Adjusted R^2	0.224	0.253	0.251	0.226	0.225

[a] Robust standard errors are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. This table uses organic-divergence samples.

refined measures and reported the results in Columns 1-2 in Table F6. All the key coefficients maintain the same direction, and some of them are still significant. There might be nuances about type I and II decision argumentation, but they are beyond the scope of the current study.

F5. Robustness Check: Collinearity Issue

Consistency and Conflict may suffer from perfect collinearity because they reflect two opposite aspects of human-algorithm decision-rule heterogeneity. However, as shown in Table F4, they are just weakly correlated (-0.319) and

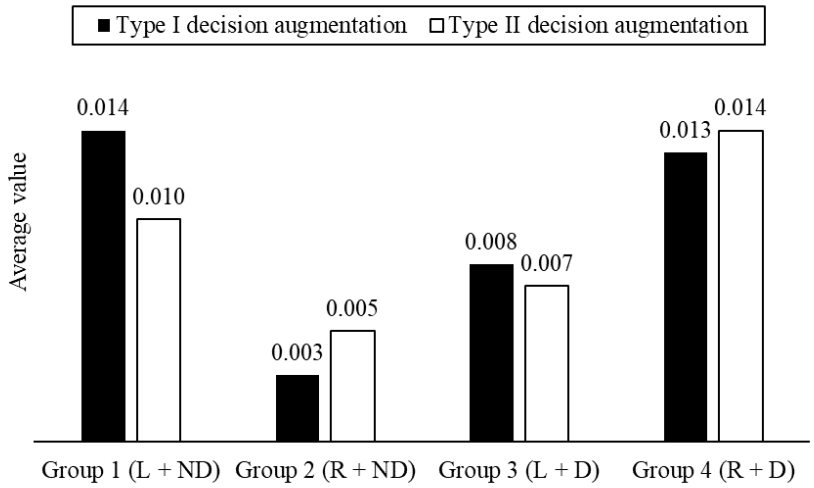


Figure F2 Type I and Type II Decision Augmentation

do not seem to bias the estimation severely. We also conducted a formal empirical test by using one variable at a time to examine the influence of the other. We reran the decision augmentation model and reported the results in Columns 3-4 in Table F6. The results are almost identical to those when both Consistency and Conflict were added to the model simultaneously.

Table F6 Multiple Robustness Checks

D.V.	Type I Decision Augmentation	Type II Decision Augmentation	Decision Augmentation	Decision Augmentation
CollaborativeDisagree × Consistency	-0.060*** (0.016)	0.016 (0.012)		-0.039** (0.017)
CollaborativeDisagree × Conflict	0.001 (0.034)	-0.029 (0.034)	0.021 (0.041)	
CollaborativeDisagree × Contradiction	0.081*** (0.012)	0.074*** (0.011)	0.141*** (0.011)	0.153*** (0.013)
CollaborativeDisagree	0.319*** (0.065)	0.087 (0.058)	0.240***(0.043)	0.377*** (0.065)
Consistency	-0.000 (0.003)	0.006 (0.004)		-0.003 (0.003)
Conflict	0.018* (0.010)	0.018** (0.009)	0.029*** (0.010)	
Contradiction	-0.001 (0.003)	0.009** (0.004)	0.010** (0.005)	0.009** (0.005)
Human Characteristics	✓	✓	✓	✓
Loan Features	✓	✓	✓	✓
Group Fixed Effects	✓	✓	✓	✓
Observations	4,451	4,451	4,451	4,451
Adjusted R^2	0.148	0.168	0.246	0.249

[a] Robust standard errors are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. This table uses organic-divergence samples.

Appendix G. Supplementary Empirical Analyses for Mechanism Examinations

G1. Antecedents of Organic Divergence

Here, we quantified the effects of different factors on organic divergence using a similar empirical strategy to that specified in Section 5.3. There were three major differences: first, the dependent variable in this analysis was the binary indicator of organic divergence; second, we estimated the model using the full sample; third, we did not include Contradiction in the regression because it was generated in the second step of human-algorithm collaboration.

Table G1 reports the results, and we only display the coefficients of the two factors of decision-rule heterogeneity and four objective factors of human characteristics. The results are mixed but still partially align with some findings in previous studies. The impact of Consistency is negative in the full sample (i.e., Column 1), which is reasonable because when human evaluators shared similar decision rules with algorithms, they were more likely to come to the same conclusion. Similarly, from the opposite direction, we observed that a higher level of Conflict led to a higher likelihood of organic divergence. Based on the results from the full sample, female evaluators and evaluators with longer working tenures were more likely to make different lending decisions from those of algorithms. However, the results in Columns 2 and 3 do not support this finding when we split the sample. On the other hand, we observed a consistent and negative effect of evaluators' previous accuracy across all samples. Considering that algorithms generally outperformed human evaluators, it is reasonable to expect a lower level of organic divergence among evaluators with higher historical accuracy rates (indicating them to be more experienced and successful). Because there is no human-algorithm collaboration involved yet for organic divergence to occur, we decided not to highlight this analysis or its implications in our study.

Table G1 Antecedents of Organic Divergence

D.V.	`OrganicDiverge`		
Sample	All groups	Groups 1&3 (Limited Info)	Groups 2&4 (Rich Info)
`Consistency`	-0.020*** (0.002)	0.012** (0.005)	0.003 (0.005)
`Conflict`	0.029*** (0.006)	0.025** (0.011)	0.005 (0.019)
`GenderEva`	0.044*** (0.008)	-0.021 (0.015)	0.014 (0.017)
`Tenure`	0.007** (0.003)	0.002 (0.006)	-0.006 (0.007)
`EducationEva`	0.003 (0.004)	-0.007 (0.007)	-0.012 (0.012)
`PreviousAccuracy`	-0.050*** (0.004)	-0.029*** (0.006)	-0.123*** (0.023)
Other Human Characteristics	✓	✓	✓
Loan Features	✓	✓	✓
Group Fixed Effects	✓	✓	✓
Observations	23,809	11,787	12,022
Adjusted R^2	0.040	0.013	0.071

[a] Robust standard errors are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. This table uses the full sample.

G2. Experimental Treatments and Collaborative Disagreement

To investigate how experimental treatments triggered the path to collaborative value, we considered both organic divergence and collaborative disagreement as dependent variables and our experimental design factors as independent variables, in addition to the control variables of human characteristics and loan features. We used three variables to reflect our two-by-two experimental design, which are RichInfo, Disclosure, and RichInfo × Disclosure. They are all dummy variables, with 1 indicating a specific experimental design factor is implemented under one scenario. Columns 1 and 3 in Table G2 indicate the predictive power of control variables. With the addition of experimental design factors (Columns 2 and 4), the predictive power increases slightly for both OrganicDivergence and Disagreement, suggesting that they are partially influenced by the experimental design. The impact of each design factor is consistent with what we found in Figure 3a. All these results confirm that organic divergence and collaborative disagreement are not easily manipulated by experimental treatments–they likely come from the distinct ways human evaluators and algorithms process information. Therefore, the mechanism centered on disagreement is likely to be general because disagreement does not directly rely on the human-algorithm collaboration task too much.

Table G2 Empirical Examination of Experimental Design and Disagreement

	OrganicDivergence		CollaborativeDisagreemnt	
	(1)	(2)	(3)	(4)
RichInfo		0.086***		-0.009*
		(0.009)		(0.005)
Disclosure		0.013		-0.020***
		(0.009)		(0.005)
RichInfo × Disclosure		-0.005		0.014*
		(0.014)		(0.007)
Human Characteristics	✓	✓	✓	✓
Loan Features	✓	✓	✓	✓
Observations	23,809	23,809	23,809	23,809
Adjusted R^2	0.028	0.033	0.012	0.013

[a] Robust standard errors are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. This table uses the full sample.

G3. Experimental Treatments and Key Factors

While we could not directly manipulate Consistency, Conflict, and Contradiction, they were nevertheless influenced by our experimental design. To investigate to what extent they were influenced, we considered them as dependent variables and our experimental treatments as independent variables, in addition to the control variables of human characteristics and loan features. We used the same three variables introduced in Appendix G2 to reflect our two-by-two experimental design. Columns 1, 3, and 5 in Table G3 indicate the predictive power of control variables. With the addition of experimental treatments (Columns 2, 4, and 6), the predictive power increases slightly for Consistency, Conflict and largely for Contradiction. The results suggest that Consistency, Conflict largely reflect

the general decision-making pattern of human evaluators, which likely stems from long-lasting habits and heuristics rather than external stimuli. Contradiction is more like a fact that was hinted at in each individual application, and it is closely related to the experimental design, especially the amount of available information in model training and the disclosure of algorithm rationale. Given the results in Table G3, we can better understand the nature of these variables and interpret the mechanism accordingly (e.g., to what extent they were stimulated by experimental treatments and to what extent experimental treatments influenced collaborative value through them).

Table G3 Empirical Examination of Experimental Design and Key Factors

	Consistency		Conflict		Contradiction	
	(1)	(2)	(3)	(4)	(5)	(6)
RichInfo		0.719***		0.180***		-0.024*
		(0.027)		(0.012)		(0.013)
Disclosure		0.531***		-0.028***		2.006***
		(0.026)		(0.010)		(0.021)
RichInfo × Disclosure		-0.141***		0.198***		1.304***
		(0.042)		(0.017)		(0.030)
Human Characteristics	✓	✓	✓	✓	✓	✓
Loan Features	✓	✓	✓	✓	✓	✓
Observations	23,809	23,809	23,809	23,809	23,809	23,809
Adjusted R^2	0.474	0.503	0.279	0.303	0.254	0.720

[a] Robust standard errors are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. This table uses the full sample.

Appendix H. Robustness Check: Focal Mechanism and Alternative Explanations

H1. A Heckman Model of the Two-stage Collaboration Process

Considering that organic divergence is the prerequisite for collaborative disagreement and that many variables influence organic divergence, we factored in the "self-selection" process of organic divergence using a Heckman two-stage model. As discussed in Appendix G1, all variables used in Table G1 were also used to predict the likelihood of organic divergence in the first stage (see Equation (H1)).

$$\begin{aligned}\mathsf{OrganicDivergence}_{\mathsf{ijg}} = \beta_0 + \beta_{11}\mathsf{Consistency}_{\mathsf{jg}} + \beta_{12}\mathsf{Conflict}_{\mathsf{jg}} + \beta_2\mathsf{HumanCharacteristics}_{\mathsf{jg}} \\ + \beta_3\mathsf{LoanFeatures}_{\mathsf{ig}} + \mathsf{Group}_{\mathsf{g}} + \varepsilon_{\mathsf{ijg}},\end{aligned} \tag{H1}$$

Controlling the influence of these variables in the first stage, we examined the impact of these variables, as well as some variables that did not influence organic divergence in the first stage, on collaborative disagreement, decision augmentation, and good decision. Equation (H2) shows the second-stage equation for collaborative disagreement while Equation (H3) shows the second-stage equation for decision augmentation and good decision.

$$\begin{aligned}\mathsf{CollaborativeDisagree}_{\mathsf{ijg}} = \alpha_0 + \alpha_{11}\mathsf{Consistency}_{\mathsf{jg}} + \alpha_{12}\mathsf{Conflict}_{\mathsf{jg}} + \alpha_2\mathsf{Contradiction}_{\mathsf{ig}} \\ + \alpha_3\mathsf{HumanCharacteristics}_{\mathsf{jg}} + \alpha_4\mathsf{LoanFeatures}_{\mathsf{ig}} + \mathsf{Group}_{\mathsf{g}} + \varepsilon_{\mathsf{ijg}},\end{aligned} \tag{H2}$$

$$\begin{aligned}\mathsf{CollaborativeValue}_{\mathsf{ijg}} = \alpha_0 + \alpha_1\mathsf{CollaborativeDisagree}_{\mathsf{ijg}} + \alpha_2\mathsf{Consistency}_{jg} + \alpha_3\mathtt{Conflict}_{jg} \\ + \alpha_4\mathsf{Contradiction}_{\mathsf{ig}} + \alpha_{51}\mathsf{CollaborativeDisagree}_{\mathsf{ijg}} \times \mathsf{Consistency}_{\mathsf{jg}} \\ + \alpha_{52}\mathsf{CollaborativeDisagree}_{\mathsf{ijg}} \times \mathsf{Conflict}_{\mathsf{jg}} \\ + \alpha_{53}\mathsf{CollaborativeDisagree}_{\mathsf{ijg}} \times \mathsf{Contradiction}_{\mathsf{ig}} + \mathsf{Controls} + \mathsf{Group}_{\mathsf{g}} + \varepsilon_{\mathsf{ijg}},\end{aligned} \tag{H3}$$

As shown in Table H1, our key findings remain similar: (1) Contradiction still increases disagreement (Column 1); (2) Disagreement on average improves decision augmentation (Column 2); (3) Disagreement is largely harmful when Consistency or Conflict is high (Columns 2 and 3); and (4) Disagreement is beneficial when Contradiction is high (Columns 2 and 3).

H2. Evaluators' Information Comprehension and Algorithm Appreciation

A potential concern about the key findings is whether human evaluators actually read and comprehended rich information. While we could not observe the number of features each evaluator paid attention to, the evaluator questionnaire provides a proxy of each evaluator's typical practice of information usage. Some of them reported that they were able to use rich information in decision-making, whereas others used rich information less frequently. Based on the median cutoff value of this information usage questionnaire item (Q15), we identified low information usage and high information usage sub-samples within the two rich-information groups (i.e., Group 3

Table H1 Robustness Check: A Heckman Model

D.V.	Collaborative Disagreement	Decision Augmentation	Good Decision
Interaction Effects			
CollaborativeDisagree × Consistency		-0.044*** (0.010)	-0.055*** (0.015)
CollaborativeDisagree × Conflict		-0.028 (0.026)	-0.105*** (0.039)
CollaborativeDisagree × Contradiction		0.155*** (0.009)	0.105*** (0.013)
CollaborativeDisagree		0.405*** (0.044)	-0.009 (0.065)
Consistency	-0.100 (0.178)	-0.011 (0.056)	0.089 (0.085)
Conflict	0.233 (0.255)	0.058 (0.079)	-0.001 (0.121)
Contradiction	0.036* (0.021)	0.009 (0.007)	-0.064*** (0.010)
Human Characteristics	✓	✓	✓
Loan Features	✓	✓	✓
Group Fixed Effects	✓	✓	✓
Observations	23,809	23,809	23,809
Wald Chi^2	22.31	1440.96	185.71
Inverse Mills Ratio	2.689 (3.635)	0.333 (1.157)	-1.264 (1.733)

[a] Robust standard errors are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. This table uses the full sample. The values of RHO for the three columns are 1.000, 0.674, and -1.000, which align with the measurement of disagreement, decision augmentation, and good decision because they were meaningful when only organic divergence occurs.

R+ND and Group 4 R+D). We reran the decision augmentation model and found the results to be largely consistent (see Columns 1 and 2 in Table H2). Regardless of evaluators' information usage levels, the direct impact of disagreement stays positive, the moderation effects of Consistency and Conflict on disagreement stay negative, and the moderation effect of Contradiction stays positive.

Similarly, evaluators may have various attitudes toward algorithms and trust their recommendations differently. Based on the median cutoff value of the algorithm appreciation questionnaire item (Q2), we split the full sample into two sub-samples, containing evaluators with low algorithm appreciation and high algorithm appreciation, respectively. We reran the decision augmentation model and observed largely consistent results (see Columns 3 and 4 in Table H2).

Table H2 Robustness Check: Human Evaluators' Information Comprehension and Algorithm Appreciation

D.V.	Decision Augmentation			
Group	Rich information		All	
Sample	Low information usage	High information usage	Low algorithm appreciation	High algorithm appreciation
CollaborativeDisagree × Consistency	-0.062 (0.040)	-0.087** (0.036)	-0.024 (0.022)	-0.056* (0.031)
CollaborativeDisagree × Conflict	-0.200* (0.115)	-0.100 (0.125)	-0.009 (0.053)	-0.247*** (0.094)
CollaborativeDisagree × Contradiction	0.163*** (0.032)	0.189*** (0.021)	0.131*** (0.014)	0.241*** (0.025)
CollaborativeDisagree	0.654*** (0209)	0.555*** (0.205)	0.390***(0.093)	0.249 (0.163)
Consistency	-0.001 (0.027)	-0.010 (0.049)	0.000 (0.007)	-0.003 (0.013)
Conflict	4.829** (2.414)	-0.027 (0.104)	-0.041 (0.025)	-0.014 (0.023)
Contradiction	0.002 (0.009)	-0.009 (0.006)	0.021*** (0.007)	-0.008 (0.005)
Human Characteristics	✓	✓	✓	✓
Loan Features	✓	✓	✓	✓
Group Fixed Effects	✓	✓	✓	✓
Observations	1,134	1,437	2,924	1,527
Adjusted R^2	0.370	0.377	0.254	0.348

[a] Robust standard errors are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. This table uses organic-divergence samples.

H3. The Importance of Algorithmic Recommendations

While the focal firm has emphasized the importance of algorithmic recommendations in the field experiment, it is unclear how seriously human evaluators took algorithmic recommendations. In other words, did they consider algorithmic recommendations as one regular piece of information or as strong suggestions for the final decision? We investigated this concern empirically by checking the influence of algorithmic recommendations on the final decision made by human evaluators. FinalApproval, a dummy variable indicating whether the final decision was approval, was used as the dependent variable. HumanOrganicApproval (where 1 means the initial, independent decision of a human evaluator was approval) and AlgorithmOrganicApproval (where 1 means the recommendation of the algorithm was approval), along with other control variables, were used to predict final approval. As shown in Table H3, the inclusion of AlgorithmOrganicApproval largely increases the adjusted R^2 of the models and reduces the coefficient of HumanOrganicApproval. The coefficients of AlgorithmOrganicApproval are also sizeable and significant. All the evidence suggested that human evaluators took algorithmic recommendations very seriously, and algorithmic recommendations were more important than any other factor, including the initial decision of human evaluators themselves.

Table H3 Robustness Check: The Importance of Algorithmic Recommendations

D.V.	FinalApproval			
HumanOrganicApproval	0.621*** (0.005)	0.621*** (0.005)	0.216*** (0.006)	0.216*** (0.006)
AlgorithmOrganicApproval			0.758*** (0.007)	0.758*** (0.007)
Consistency		0.007*** (0.002)		-0.001 (0.001)
Conflict		0.018*** (0.005)		0.009*** (0.003)
Human Characteristics	✓	✓	✓	✓
Evaluator Survey		✓		✓
Loan Features	✓	✓	✓	✓
Group Fixed Effects	✓	✓	✓	✓
Observations	23,809	23,809	23,809	23,809
Adjusted R^2	0.555	0.556	0.872	0.872

[a] Robust standard errors are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. This table uses the full sample.

Appendix I. Lab Experiment Design and Analysis

To determine if our main findings hold in other scenarios, we conducted a lab experiment. The motivation was to replicate the field experiment design and examine the robustness of the performance comparison results (practical generalizability) and the mechanism (theoretical generalizability). The experimental context, design, process, and variable constructions are reported as follows.

I1. Experimental Context

The lab experiment focused on predicting students' academic performance in math. We selected this particular setting for several reasons:

1. **Relevance of task**: Predicting students' academic performance is a widespread real-world practice. It represents a typical example of performance prediction tasks, which is applicable to scenarios such as student admission, grant allocation, and employee recruitment.
2. **Numerical prediction outcome**: Unlike the binary outcome in our field experiment, the outcome of this academic performance prediction task is numerical. This variation allows us to test the extendibility of our findings to different types of evaluation outcomes.
3. **Moderate complexity level**: The prediction of academic performance is influenced by a variety of factors and can be handled by ordinary people without extensive training or years of experience (as long as they have some education history). In this task, some decision factors are observable and quantifiable, such as demographics and assignment completion, which can be effectively processed by ML algorithms. Others, however, are based on subjective human insights and thus are difficult to capture fully by algorithms. This complexity level suggests that a collaborative approach between humans and algorithms might enhance overall prediction accuracy, potentially surpassing the performance of either humans or algorithms working independently.

Specifically, we attended to the prediction of mathematics grades of secondary students at two Portuguese schools, using a public dataset from the UCI Machine Learning Repository (`https://archive.ics.uci.edu/ml/datasets/Student+Performance`). This dataset includes 395 student cases, each with 30 attributes such as gender, age, past class failures, and final mathematics grades.[6]

I2. Experimental Design

Following the same approach used in our field experiment, we devised a two-by-two experimental design to represent four collaboration scenarios. Before the experiment, we trained a Decision Tree Regression to predict student

[6] The raw final mathematics grades, which range from 0 to 20, were aggregated into five levels for our experiment in ML training and performance prediction: Level 0 for a grade of 0, Level 1 for grades 1 to 5, Level 2 for grades 6 to 10, Level 3 for grades 11 to 15, and Level 4 for grades 16 to 20.

mathematics grades under both the limited- and rich-information scenarios. [7] Under the limited-information scenario, following You et al. (2021), we identified 10 features that included students' gender, age, address, family size, student's guardian, reason to choose this school, number of past class failures (*failures*), number of school absences (*absences*), weekly study time (*studytime*), and free time after school (*freetime*). These features are often considered intuitive indicators of student mathematics performance. Under the rich-information scenario, we expanded the variable set by including an additional 20 features, such as the educational backgrounds of the student's mothers and fathers, travel time from home to school, involvement in extracurricular activities, internet access at home, workday alcohol consumption, frequency of social outings with friends (*goout*), and availability of extra educational support at school (*schoolsup*).

Important features used by the algorithm					
	Avg value of students obtaining grade level 0	Avg value of students obtaining grade level 1	Avg value of students obtaining grade level 2	Avg value of students obtaining grade level 3	Avg value of students obtaining grade level 4
absences (–)	0	12.75	7.49	5.51	4.30
failures (–)	0.92	1.13	0.47	0.12	0.03
age (–)	17.08	16.75	16.86	16.54	16.40
freetime (–)	3.13	4.13	3.21	3.21	3.35
studytime (+)	1.97	1.50	1.96	2.09	2.20
Notes. (+) means the feature is positively related to high grade level, (–) means the feature is negatively related to high grade level. The statistics in each cell were derived from the original student grade data (i.e., the fact).					

(a) **Limited-Information Scenario**

Important features used by the algorithm					
	Avg value of students obtaining grade level 0	Avg value of students obtaining grade level 1	Avg value of students obtaining grade level 2	Avg value of students obtaining grade level 3	Avg value of students obtaining grade level 4
absences (–)	0	12.75	7.49	5.51	4.30
failures (–)	0.92	1.13	0.47	0.12	0.03
goout (–)	3.21	4.13	3.26	2.95	2.95
schoolsup (–)	0.03	0.25	0.21	0.10	0.03
studytime (+)	1.97	1.50	1.96	2.09	2.20
Notes. (+) means the feature is positively related to high grade level, (–) means the feature is negatively related to high grade level. The statistics in each cell were derived from the original student grade data (i.e., the fact).					

(b) **Rich-Information Scenario**

[a] Feature importance was based on SHAP values computed from the Decision Tree Regression model. The top five important features were displayed because they collectively contributed to predicting student math grades.
[b] These important features were fixed for all cases assigned in the same group.

Figure I1 Top Five Important Features Used by Algorithms Disclosed in the Lab Experiment

[7] We also attempted XGBoost, but it completely outperformed human evaluators. Given the implicit prerequisite for collaborative value is that humans and algorithms have their own advantages, we purposely did not select a close-to-perfect algorithm in this experiment.

The model training process used the complete set of 395 cases and employed 5-fold cross-validation. The average Mean Squared Errors (MSEs) were 1.072 for the limited-information scenario and 0.946 for the rich-information scenario. The improved prediction performance in the rich-information scenario highlights the additional value of the extra 20 features over the basic features. We then recruited students from a U.S. university as experiment participants and randomly divided them into four experimental groups. Each participant was first assigned 30 randomly-selected student cases and tasked with predicting the students' mathematics grade levels (ranging from 0 to 4). In the second step, we presented the participants with algorithmic recommendations and algorithm rationale (in the disclosure groups only), prompting them to make final predictions. We disclosed the top five important features used by the algorithms in both limited-information and rich-information scenarios.

As a result of the experimental design and execution procedure, we ended up with four experimental groups, differentiated by information volume (**L**imited vs. **R**ich) and disclosure of algorithm rationale (**N**on-**D**isclosure vs. **D**isclosure): **Group 1 (L+ND)**, **Group 2 (R+ND)**, **Group 3 (L+D)**, and **Group 4 (R+D)**. These definitions are consistent with those used in our field experiment. Figure I1 illustrates the disclosure of algorithm rationale for the limited- and rich-information scenarios used in Groups 3 and 4.

I3. Experimental Sample, Data, and Variable Constructions

We conducted our lab experiment at a research university, and all participants were university students. Given the focus on math grade prediction, university students were deemed suitable participants due to their familiarity with academic settings. The experiment received approval from the university's Institutional Review Board (IRB). We recruited 40 participants per experimental group and collected 30 cases for each participant, totaling 4,800 observations from the lab experiment. Table I1 reports the randomization check results for the lab experiment. Overall, the differences in student case attributes and experimental participant backgrounds across groups are insignificant, with all p-values from the between-group F-tests exceeding 0.1, indicating successful randomization.

We followed the approach used in our field experiment to generate key variables. Because the outcome variable is of numerical value (i.e., from 0 to 4), we slightly adjusted the measurement of decision augmentation. Detailed measurements and statistics of key variables are reported in Table I2.

Table I1 Randomization Check for the Lab Experiment

	(Major) student case attributes					
Group	Age	Absences	Failures	Freetime	Studytime	Schoolsup
1. L+ND	16.700	4.233	0.533	3.100	2.000	0.067
	(1.643)	(4.981)	(0.937)	(1.185)	(0.983)	(0.254)
2. R+ND	16.667	3.833	0.467	3.333	1.833	0.067
	(1.605)	(3.455)	(0.973)	(1.295)	(0.950)	(0.254)
3. L+D	17.333	4.233	0.467	3.567	2.133	0.067
	(1.213)	(7.459)	(0.937)	(1.305)	(0.973)	(0.254)
4. R+D	17.433	3.967	0.567	3.300	2.267	0.067
	(1.251)	(3.943)	(1.006)	(1.119)	(0.980)	(0.254)

	Experimental participant background		
Group	Participant gender	Participant age	Participant years working
1. L+ND	0.400	25.925	2.644
	(0.496)	(2.859)	(2.579)
2. R+ND	0.400	25.800	2.444
	(0.496)	(3.560)	(2.675)
3. L+D	0.400	25.925	2.644
	(0.496)	(2.616)	(2.815)
4. R+D	0.400	25.800	2.444
	(0.496)	(3.340)	(2.655)

[a] The 30 cases reviewed by each participant within a group are the same. So the statistic values in the table are based on these 30 unique cases for each group. Standard deviations are reported in parentheses.
[b] L (R) = limited (rich) information; ND (D) = without (with) disclosure of important features used by algorithms.
[c] Age: the student's age (numeric: from 15 to 22). absences: number of school absences (numeric: from 0 to 93). Failures: number of past class failures (numeric: from 1 to 4). Freetime: free time after school (numeric: from 1 = very low to 5 = very high). Studytime: weekly study time (numeric: 1 = less than 2 hours, 2 = 2 to 5 hours, 3 = 5 to 10 hours, or 4 = more than 10 hours). Schoolsup: extra educational support (binary: 1 = yes, 0 = no).
[d] Participant gender: the experimental participant's gender (binary: 1 = female, 0 = male). Participant age: Participant's age (numeric: from 23 to 39). Participant years working: the participant's total years of work experience (numeric: from 0 to 15.75).

Table I2 Measurements and Descriptive Statistics of Key Variables in the Lab Experiment

Variable	Definition	Min	Mean	Median	Max
Collaborative Performance					
GoodDecision	Binary; =1 if the final predicted grade aligns with the true grade.	0	0.543	1	1
Augmentation	=1 if the final predicted grade is closer to the true grade than is the algorithm-only predicted grade; =-1 if the opposite; =0 if the same.	-1	0.001	0	1
Decision Inconsistency					
OrganicDiverge	Binary; =1 if a human's initial grade prediction is different from the algorithmic prediction.	0	0.135	0	1
CollaborativeDisagree	Binary; =1 if human does not follow the algorithmic recommendation.	0	0.049	0	1
Decision-rule Heterogeneity					
Consistency	# of significant student features agreed by humans and algorithms.	0	1.719	2	4
Conflict	# of significant student features considered in opposite directions by humans and algorithms.	0	0.413	0	1
Perceived algorithm quality					
Contradiction	# of important features used by algorithms exhibiting contradictory logic. Its value is zero for applications in non-disclosure groups.	0	0.167	0	3

Statistics for all variables are based on the full sample.

References

Allen, Ryan, Prithwiraj Choudhury. 2022. Algorithm-augmented work and domain experience: The countervailing forces of ability and aversion. *Organization Science* **33**(1) 149–169.

Bauer, Kevin, Moritz von Zahn, Oliver Hinz. 2023. Expl (ai) ned: The impact of explainable artificial intelligence on users' information processing. *Information Systems Research* .

Cao, Sean, Wei Jiang, Junbo L Wang, Baozhong Yang. 2021. From man vs. machine to man+ machine: The art and ai of stock analyses. Tech. rep., National Bureau of Economic Research.

Castelo, Noah, Maarten W Bos, Donald R Lehmann. 2019. Task-dependent algorithm aversion. *Journal of Marketing Research* **56**(5) 809–825.

Chen, Valerie, Q Vera Liao, Jennifer Wortman Vaughan, Gagan Bansal. 2023. Understanding the role of human intuition on reliance in human-ai decision-making with explanations. *arXiv preprint arXiv:2301.07255* .

Choudhury, Prithwiraj, Evan Starr, Rajshree Agarwal. 2020. Machine learning and human capital complementarities: Experimental evidence on bias mitigation. *Strategic Management Journal* **41**(8) 1381–1411.

Dietvorst, Berkeley J, Joseph P Simmons, Cade Massey. 2015. Algorithm aversion: people erroneously avoid algorithms after seeing them err. *Journal of Experimental Psychology: General* **144**(1) 114.

Fu, Runshan, Yan Huang, Param Vir Singh. 2021. Crowds, lending, machine, and bias. *Information Systems Research* **32**(1) 72–92.

Fügener, Andreas, Jörn Grahl, Alok Gupta, Wolfgang Ketter. 2021. Will humans-in-the-loop become borgs? merits and pitfalls of working with ai. *Management Information Systems Quarterly (MISQ)-Vol* **45**.

Fügener, Andreas, Jörn Grahl, Alok Gupta, Wolfgang Ketter. 2022. Cognitive challenges in human–artificial intelligence collaboration: investigating the path toward productive delegation. *Information Systems Research* **33**(2) 678–696.

Ge, Ruyi, Zhiqiang Zheng, Xuan Tian, Li Liao. 2021. Human–robot interaction: when investors adjust the usage of robo-advisors in peer-to-peer lending. *Information Systems Research* **32**(3) 774–785.

Ibrahim, Rouba, Song-Hee Kim, Jordan Tong. 2021. Eliciting human judgment for prediction algorithms. *Management Science* **67**(4) 2314–2325.

Lehmann, Cedric A, Christiane B Haubitz, Andreas Fügener, Ulrich W Thonemann. 2022. The risk of algorithm transparency: How algorithm complexity drives the effects on the use of advice. *Production and Operations Management* **31**(9) 3419–3434.

Liu, Meng, Xiaocheng Tang, Siyuan Xia, Shuo Zhang, Yuting Zhu, Qianying Meng. 2023. Algorithm aversion: Evidence from ridesharing drivers. *Management Science* .

Lu, Tian, Beibei Li, Yingjie Zhang. 2023. Profit vs. equality? the case of financial risk assessment and a new perspective on alternative data. *MIS Quarterly* **47**(4) 1517–1556.

Lu, Tian, Xianghua Lu, Yiyu Huang, Hai Wang. 2024. Augmented algorithms, adaptive humans? evidence from a natural experiment. *SSRN 4298793* .

Lu, Tian, Yingjie Zhang. 2024. 1+ 1¿ 2? information, humans, and machines. *Information Systems Research* .

Luo, Xueming, Marco Shaojun Qin, Zheng Fang, Zhe Qu. 2021. Artificial intelligence coaches for sales agents: Caveats and solutions. *Journal of Marketing* **85**(2) 14–32.

Sun, Jiankun, Dennis J Zhang, Haoyuan Hu, Jan A Van Mieghem. 2022. Predicting human discretion to adjust algorithmic prescription: A large-scale field experiment in warehouse operations. *Management Science* **68**(2) 846–865.

Te'eni, Dov, Inbal Yahav, Alexely Zagalsky, David Schwartz, Gahl Silverman, Daniel Cohen, Yossi Mann, Dafna Lewinsky. 2023. Reciprocal human-machine learning: A theory and an instantiation for the case of message classification. *Management Science* .

von Walter, Benjamin, Dietmar Kremmel, Bruno Jäger. 2022. The impact of lay beliefs about ai on adoption of algorithmic advice. *Marketing Letters* **33**(1) 143–155.

You, Sangseok, Cathy Liu Yang, Xitong Li. 2022. Algorithmic versus human advice: does presenting prediction performance matter for algorithm appreciation? *Journal of Management Information Systems* **39**(2) 336–365.

You, Sukkyung, Eui Kyung Kim, Sun Ah Lim, Myley Dang. 2021. Student and teacher characteristics on student math achievement. *Journal of Pacific Rim Psychology* **15** 1834490921991428.

Zhou, Jilei, Cong Wang, Fei Ren, Guoqing Chen. 2021. Inferring multi-stage risk for online consumer credit services: An integrated scheme using data augmentation and model enhancement. *Decision Support Systems* **149** 113611.